Transfer Learning and CNN Applications

Scaler School of Technology | Deep Learning I | Week 5 — Session 1/2 | May 2026

What Is This Lecture About?

You've built CNN architectures from scratch — now it's time to use them **without** training from scratch. This session covers transfer learning (the technique behind 90%+ of real-world deep learning), then extends CNNs beyond classification to object detection and image segmentation.

1 Transfer Learning — Feature Extraction & Fine-Tuning

Why train a CNN from scratch when someone already trained one on millions of images? Learn to reuse pre-trained models as feature extractors or fine-tune them for your specific task with minimal data. *Like a chef trained in French cuisine adapting to Italian — the knife skills transfer, only the flavors change.*

2 Object Detection — YOLO & Faster R-CNN

Classification says "there's a cat." Detection says "there's a cat **here** and a dog **there**." We'll trace the evolution from naive sliding windows to modern one-shot detectors that process video in real-time. *Like going from "someone's in the room" to "Alice is by the window and Bob is at the desk."*

3 Image Segmentation — FCN & U-Net

Detection gives bounding boxes; segmentation classifies **every single pixel**. We'll see how encoder-decoder architectures with skip connections produce pixel-perfect masks — critical for medical imaging, self-driving cars, and satellite analysis.

What to Expect in the Session

- Visualizing what pre-trained CNN layers have learned
- ⋆ Hands-on: **feature extraction vs. fine-tuning** on a small dataset
- Running a pre-trained object detector on real images
- Seeing segmentation masks produced by DeepLabV3
- 2 quizzes on transfer learning strategy decisions
- Decision framework: which approach for which scenario

How to Prepare (Optional but Helpful)

- ✓ **Recall** the CNN architecture hierarchy: LeNet → AlexNet → VGG → ResNet (Session 8)
- ✓ **Recall** what skip connections do and why ResNet works (Session 8)
- ✓ **Think about:** "If you only have 500 labeled images, how would you still use a model designed for 1.2M images?"
- ✓ **Skim** `torchvision.models` docs if curious about what pre-trained models are available

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Duration: ~2 hours **Prerequisites:** CNN architectures, ResNet skip connections (Sessions 7–8) **Instructor:** Priyansh Saxena